C8 Chemical Analysis – Knowledge Organiser

AQA GCSE Chemistry • Set 1: Purity, Formulations and Chromatography **Paper 2**

#	Question	Answer
1	Pure substance (chemistry)	Single element or compound
2	Melting point of a pure substance	Sharp, specific value
3	Melting point of an impure substance	Lower and over a range
4	Formulation	Mixture designed as a useful product
5	Examples of formulations	Fuels, paints, medicines, alloys, fertilisers
6	Two phases in chromatography	Stationary and mobile
7	R_f equation	Distance moved by substance ÷ distance moved by solvent
8	Pure substance on a chromatogram	One spot
9	Why the start line is drawn in pencil	Pencil does not dissolve
10	How a substance is identified	Same R_f as a known sample

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AQA GCSE Chemistry • Set 2: Gas Tests **Paper 2**

#	Question	Answer
1	Test for hydrogen	Lighted splint: squeaky pop
2	Test for oxygen	Glowing splint relights
3	Test for carbon dioxide	Limewater turns milky
4	Test for chlorine	Damp litmus paper is bleached
5	Limewater	Calcium hydroxide solution
6	Range of R_f values	Between 0 and 1
7	Why the solvent is below the start line	Spots would dissolve into the solvent
8	Stationary phase in paper chromatography	Paper
9	Mobile phase in paper chromatography	Solvent
10	Why a spot stays on the start line	Insoluble in the solvent

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AQA GCSE Chemistry • Set 3: Flame Tests and Hydroxides Paper 2 Triple only

#	Question	Answer
1	Flame colour: lithium	Crimson
2	Flame colour: sodium	Yellow
3	Flame colour: potassium	Lilac
4	Flame colour: calcium	Orange-red
5	Flame colour: copper	Green
6	Al^{3+} with NaOH	White precipitate, dissolves in excess
7	Ca^{2+} with NaOH	White precipitate
8	Mg^{2+} with NaOH	White precipitate
9	Cu^{2+} with NaOH	Blue precipitate
10	Fe^{2+} with NaOH	Green precipitate

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AQA GCSE Chemistry • Set 4: Negative Ions and Instrumental Methods Paper 2 Triple only

#	Question	Answer
1	Fe^{3+} with NaOH	Brown precipitate
2	Test for carbonate ions	Dilute acid; gas turns limewater milky
3	Reagents to test for halide ions	Dilute nitric acid + silver nitrate
4	Chloride with silver nitrate	White precipitate
5	Bromide with silver nitrate	Cream precipitate
6	Iodide with silver nitrate	Yellow precipitate
7	Test for sulfate ions	Dilute HCl + barium chloride: white precipitate
8	Advantages of instrumental methods	Accurate, sensitive, rapid
9	Flame emission spectroscopy identifies ions by	Wavelengths of light emitted (line spectrum)
10	Why flame tests fail for mixtures	Some colours mask others